First-Order ODEs (Ch 1)

1. Linear. $x' + P(t)\,x = Q(t)$

$$\mu = e^{\int P\,dt}, \quad x = \frac{1}{\mu}\left[\int \mu\,Q\,dt + C\right]$$

2. Separable.

$$\frac{dx}{dt} = f(t)\,g(x) \;\Rightarrow\; \int \frac{dx}{g(x)} = \int f(t)\,dt$$

3. Bernoulli. $x' + P(t)\,x = Q(t)\,x^n, \quad n \neq 0, 1$
Sub $y = x^{1-n}$, $\quad y' = (1-n)x^{-n}x'$. Divide through by x^n, get linear in y.

4. Riccati. $x' = p(t) + q(t)\,x + r(t)\,x^2$
Need one particular solution x_p. Sub $x = x_p + u$, get Bernoulli in u.
Particular solution: guess at^c where $a, c \in \mathbb{R}$.

5. Exact. $M(x,y)\,dx + N(x,y)\,dy = 0$
Test: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Find F: $\frac{\partial F}{\partial x} = M$, $\frac{\partial F}{\partial y} = N$, solution is $F = C$.

6. Homogeneous. $y' = F(y/x)$
Sub $v = y/x$, so $y = vx$, $y' = v + xv'$. Becomes separable in v.

Lipschitz Continuity (Ch 2)

$|f(x) - f(y)| \leq L\,|x - y|$ for all $x, y \in D$
If f differentiable and $|f'| \leq L$ everywhere, then f is Lipschitz with constant L (by MVT).
Lipschitz $\Rightarrow$ unif. cont. $\Rightarrow$ continuous

Matrix Exponential (Ch 3)

$e^{tA} = I + tA + \frac{t^2}{2!}A^2 + \cdots$
$A = PDP^{-1} \Rightarrow e^{tA} = Pe^{tD}P^{-1}$
Nilpotent: $A^2 = 0 \Rightarrow e^{tA} = I + tA$
Decomposition: $A = \lambda I + N$, $\; e^{tA} = e^{t\lambda}e^{tN}$
If $(A - \lambda I)^2 = 0$: $\; e^{tA} = e^{t\lambda}[I + t(A - \lambda I)]$

2×2 *distinct eigenvalues.* $e^{tA} = \frac{e^{t\lambda_1}(A - \lambda_2 I) - e^{t\lambda_2}(A - \lambda_1 I)}{\lambda_1 - \lambda_2}$

Linear Systems (Ch 3–4)

$\mathbf{x}(t) = e^{tA}\,\mathbf{x}(0)$
Eigenvalues: $\lambda = \frac{\mathrm{tr} \pm \sqrt{\mathrm{tr}^2 - 4\det}}{2}$
Duhamel: $\mathbf{x}(t) = e^{tA}\mathbf{x}(0) + \int_0^t e^{(t-s)A}\mathbf{b}(s)\,ds$

Picard Iteration

$x_{n+1}(t) = x_0 + \int_{t_0}^t f(s,\, x_n(s))\,ds$
Start $x_0(t) = x_0$. Iterate, recognize pattern.

Second-Order ODEs

$ar^2 + br + c = 0$
Distinct: $C_1e^{r_1 t} + C_2e^{r_2 t}$ Repeated: $(C_1 + C_2 t)e^{rt}$
Complex $\alpha \pm \beta i$: $e^{\alpha t}(C_1 \cos\beta t + C_2 \sin\beta t)$

Undetermined coefficients. For $ax'' + bx' + cx = g(t)$, guess x_p based on $g(t)$:
$g(t) = ke^{\alpha t} \quad \to \quad$ try $Ae^{\alpha t}$
$g(t) = k\sin(\beta t)$ or $k\cos(\beta t) \quad \to \quad$ try $A\cos\beta t + B\sin\beta t$
$g(t) = kt^n$ (poly deg n) $\quad \to \quad$ try $A_n t^n + A_{n-1}t^{n-1} + \cdots + A_0$
$g(t) = ke^{\alpha t}\sin(\beta t) \quad \to \quad$ try $e^{\alpha t}(A\cos\beta t + B\sin\beta t)$
$g(t) = kt^n e^{\alpha t} \quad \to \quad$ try $(A_n t^n + \cdots + A_0)\,e^{\alpha t}$
$g(t) =$ sum $\quad \to \quad$ guess is sum of individual guesses

Modification rule. If any term in your guess already solves the homogeneous equation, multiply the entire guess by t. If it still overlaps, multiply by t^2.

Cauchy–Euler. $t^2x'' + atx' + bx = g(t)$. Sub $x = t^r$, solve $r(r-1) + ar + b = 0$. Distinct $r_1 \neq r_2$: $C_1t^{r_1} + C_2t^{r_2}$. Repeated: $(C_1 + C_2\ln|t|)\,t^r$.

Variation of parameters. $x_p = u_1x_1 + u_2x_2, \quad u_1' = \frac{-x_2\,g/a}{W}$, $u_2' = \frac{x_1\,g/a}{W}, \quad W = x_1x_2' - x_2x_1'$

Metric Spaces & Supremum Distance

A metric on a set X is a function $d : X \times X \to \mathbb{R}$ satisfying:
$d(f,g) \geq 0$ and $d(f,g) = 0 \Leftrightarrow f = g$
$d(f,g) = d(g,f)$
$d(f,h) \leq d(f,g) + d(g,h)$

Supremum distance. For continuous $f, g : [a,b] \to \mathbb{R}$:
$d_\infty(f,g) = \sup_{x\in[a,b]} |f(x) - g(x)| = \|f - g\|_\infty$

Computing d_∞. Let $h(x) = f(x) - g(x)$, find critical points via $h'(x) = 0$, evaluate $|h|$ at critical points and endpoints a, b, take the largest value.

Completeness. $(C[a,b],\, d_\infty)$ is a complete metric space: every Cauchy sequence converges to a limit in $C[a,b]$.

L^2 Inner Product Space

$f \in L^2[a,b]$ if $\displaystyle\int_a^b |f(x)|^2\,dx < \infty$

Inner product: $\langle f,\, g\rangle = \displaystyle\int_a^b f(x)\,g(x)\,dx$

Norm: $\|f\| = \sqrt{\langle f,\, f\rangle} = \sqrt{\displaystyle\int_a^b |f|^2\,dx}$

Orthogonality: $\langle f,\, g\rangle = 0$
Cauchy–Schwarz: $|\langle f,\, g\rangle| \leq \|f\|\,\|g\|$

Showing $f \in L^2$. Compute $\int_a^b |f|^2\,dx$ and show it's finite.
Bounded shortcut: $|f| \leq M$ on $[a,b] \Rightarrow \int |f|^2 \leq M^2(b-a) < \infty$

Trig Orthogonality on $[-\pi, \pi]$

$$\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = \pi\delta_{mn} \quad (m,n \geq 1)$$
$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = \pi\delta_{mn} \quad (m,n \geq 1)$$
$$\int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,dx = 0 \quad \text{for all } m,n$$
$$\int_{-\pi}^{\pi} 1\cdot\cos(nx)\,dx = 0 \qquad \int_{-\pi}^{\pi} 1\cdot\sin(nx)\,dx = 0$$
$$\int_{-\pi}^{\pi} 1\,dx = 2\pi$$

Same structure on $[0, 2\pi]$ with identical values.

Best L^2 Approximation

Given orthogonal set $\{\phi_0, \ldots, \phi_n\}$, the best approximation of f:

$$\hat{f} = \sum_{k=0}^{n} c_k\,\phi_k, \qquad c_k = \frac{\langle f,\, \phi_k\rangle}{\langle \phi_k,\, \phi_k\rangle}$$

"Best" means minimizes $\|f - \hat{f}\|_{L^2}$.

For $\{1, \cos x, \sin x\}$ on $[-\pi, \pi]$.
$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$
$$c_1 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos x\,dx$$
$$c_2 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin x\,dx$$

Even/odd shortcuts. f even $\Rightarrow f\cdot\sin$ is odd $\Rightarrow c_2 = 0$
f odd $\Rightarrow c_0 = 0,\ c_1 = 0$

Integral Reference

$\int e^{at}dt = \frac{e^{at}}{a} \qquad \int te^{at}dt = \frac{e^{at}(at-1)}{a^2}$
$\int t^2e^{at}dt = e^{at}\left(\frac{t^2}{a} - \frac{2t}{a^2} + \frac{2}{a^3}\right)$
$\int te^{t^2}dt = \frac{1}{2}e^{t^2} \qquad \int t^n dt = \frac{t^{n+1}}{n+1}$
$\int x^2\cos x\,dx = (x^2-2)\sin x + 2x\cos x$
$\int x^2\sin x\,dx = (2-x^2)\cos x + 2x\sin x$
$\int u\,dv = uv - \int v\,du$

Other

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$e^{a+bi} = e^a(\cos b + i\sin b)$

$\cos^2 x = \frac{1+\cos 2x}{2} \qquad \sin^2 x = \frac{1-\cos 2x}{2}$

$|\sin x| \leq 1 \qquad |\cos x| \leq 1$

MVT: $\exists\, c$: $f'(c) = \frac{f(b)-f(a)}{b-a}$